WEEK 5 · DAY 1 · RULE BOOK

No Expo! Honest Train/Test Splits + Quiz 3

Follow it step by step. Every common error and its fix is here.

Your goal today: catch a model cheating, build an honest split by time, then referee a fair fight between the isolation forest and the Week 4 control chart. **14 green ✅ ticks** to collect. Time: about 45 minutes, then Quiz 3. Keep this page open next to Colab.

Before you start (2 minutes)

- ☐ Open `W5D1-exercise` in Colab → **File** → **Save a copy in Drive**.
- ☐ Run **STEP 0**. Wait for ✅ `Ready!`
- ☐ Run the three ▶️ cells under STEP 0 (data, scorer, Week 4 tools). Everything below needs them.
- ☐ Run every cell in order. Each ✍️ cell depends on the cells above it.

What "expo" looks like in time series

Expo	What it looks like	The honest version
👥 **Neighbours in training**	A random split: the test hour at 10:00 has 09:00 and 11:00 in training	Split **by time**: past to train, future to test
🔮 **Features that peek ahead**	`shift(-1)` , `rolling(..., center=True)`	Only `shift(1)` , `shift(24)` , plain `rolling(24)` : the **14:00 test**
🔑 **Settings tuned on the key**	Picking `k` or `contamination` by looking at the test answers	Fix settings first, or tune on the **training** months only

The 4 rules of a fair fight (write them down BEFORE you score)

#	Rule	Today
1	Same held-out months	2 Sep – 31 Dec
2	Same hours	steady sunlight (`steady`)
3	Same scorer	`score_events` (one flag inside = caught, outside = false alarm)
4	Settings fixed first	chart `k = 3` , forest `contamination = 0.005`

The exercise, step by step

Task	What you do	The tool you need
A1	Random split: share of fault hours the copycat found	`pred[y_te == 1].mean()`
A2	Split by time: same question	`CUT = "2025-09-01"`
A3	Which event did it still catch?	read the table, e.g. `"EV05"`
B1	Build train, embargo, test	`TRAIN_END = "2025-09-01"`, `TEST_START = "2025-09-02"`
B2	How long must the embargo be?	the longest window, in hours
C1	Contender 1: Week 4 chart, sunlit hours	`K = 3`
C2	Real catch or lucky catch?	`"real"` or `"lucky"`
C3	Contender 2: same chart, steady hours	`CONTEXT = steady` (no quotes)
C4	Contender 3: the forest on the lean features	`CONTAMINATION = 0.005`
C5	Contender 4: the chart on the lean features	`K = 3`
C6	Your recommendation	1, 2, 3 or 4
D1	Rolling origin: four exams	`GAP = 24`
D2	A dirty reference period	`DIRTY_FIT = steady` (no quotes)
🌶️ **Bonus**	Tune the forest honestly, on the past only	the smallest budget that catches all 3 past events

The code you'll use

```
train = bus.index <  "2025-09-01"          # every hour BEFORE 1 Sep
test  = bus.index >= "2025-09-02"          # every hour FROM 2 Sep (1 Sep = embargo)
assert bus.index[train].max() < bus.index[test].min()

chart_flags(X, train & steady, steady)             # 3-sigma chart: fit on train, judge
steady hours
forest_flags(X, train & steady, steady)            # isolation forest, same rows
score_events(flags[test], labels)                  # score the test months only
event_table(flags[test], labels)                   # caught? how late? how many hours
lit?
```

`train & steady` means "training months **and** steady sunlight". The first argument says which rows the detector **learns** from; the second says which rows it **judges**.

When things go wrong: error → fix

You see	It means	Fix
`AssertionError: ✍️ Fill in … first`	A `___` is still there	Replace it, run the cell again
`AssertionError: ✍️ Finish … first`	You skipped an earlier task	Do that task, then come back
`AssertionError: ✍️ Use the variable itself, without quotes`	You typed `"steady"`	Type `steady` (it's a variable, not text)
`NameError: name 'chart_flags' …` / `'X'` / `'steady'`	You skipped the ▶️ tools cell	Run the three ▶️ cells under STEP 0
`NameError: name 'train' is not defined`	You skipped B1	Do B1 first
`AssertionError: leakage: training overlaps the test`	Your dates are the wrong way round	`TRAIN_END` must come **before** `TEST_START`
`SyntaxError: leading zeros in decimal integer literals …`	You typed a date without quotes	Dates are text: `"2025-09-01"`
Your forest's false alarms differ a little from your neighbour's	Different `random_state` or library version	That's the forest's dice. The check accepts a range
Everything is weird	–	**Runtime → Restart session and run all**

Rules of thumb you'll keep forever

1. **Never shuffle time series** before splitting. Train on the past, test on the future.
2. **Embargo ≥ your longest feature window.** Then `assert` the split.
3. **Fix your settings before you look at the test.** Tune only on the training months.
4. **Same months, same hours, same scorer** for every detector you compare.
5. **Ask why a flag fired.** A catch for the wrong reason is a lucky catch.
6. **If the simple model wins or ties, recommend it.** That's a professional result.
7. **Write down your reference period** and why operations believe it was quiet.

🏁 **Finished?** Run `scoreboard()`. Then try the 🌶️ honest-tuning task, and get ready for **Quiz 3** (phones down, no AI, 20 minutes).